

Analyzing the Performance of Hybrid PSO for Optimal Directional Overcurrent Relay Coordination with Different IEC Characteristics in the IEEE 6-Bus System

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ABSTRACT

The coordination of Directional Overcurrent Relays (DOCRs) is a fundamental task in power system protection, ensuring reliable fault clearance and system stability. However, the nonlinear nature of relay characteristics and the large number of coordination constraints make this a complex optimization problem. This paper presents a comparative study between traditional Particle Swarm Optimization (PSO) and Hybrid PSO (HPSO), which integrates PSO with Simulated Annealing (SA), for the optimal coordination of DOCRs in the IEEE 6-bus system. Three IEC inverse-time relay characteristics—Standard Inverse (SI), Very Inverse (VI), and Extremely Inverse (EI)—are investigated to evaluate their impact on relay coordination with mention methods. The objective function is formulated to minimize the total operating time of primary relays while satisfying selectivity constraints. Simulation results demonstrate that HPSO consistently achieves superior optimization performance, yielding lower objective function values and faster convergence compared to conventional PSO in coordination problems with different IEC operational characteristics. The findings confirm the effectiveness of HPSO in enhancing the robustness and efficiency of relay coordination under different IEC characteristics.

Keywords

Directional Overcurrent Relay (DOCR), Relay Coordination, Particle Swarm Optimization (PSO), Hybrid PSO (HPSO), Simulated Annealing (SA), IEC Inverse-Time Characteristics, Power System Protection, IEEE 6-Bus System.

1. INTRODUCTION

The reliability and security of modern power systems strongly depend on the proper operation of protection devices. These devices are designed to detect electrical faults in the system and isolate the faulty section by sending trip signals to equipment such as circuit breakers or disconnectors. Among these devices, directional overcurrent relays (DOCRs) are widely employed due to their simplicity, cost-effectiveness, and ability to provide backup protection or main protection specifically in low voltage distribution systems. To enhance the reliability of electrical systems, additional relays are integrated to serve as a second line of defense in case the primary protection fails, with a specific time sequence known as relay coordination [1, 2]. Relay coordination refers to the systematic adjustment of protection devices so that the primary relay operates first, while backup relays intervene only if the primary protection fails, thereby ensuring both the reliability and stability of the power system [3]. However, when multiple relays are connected in a meshed network, miscoordination may occur if relays operate simultaneously or in the wrong order. Therefore, the optimal coordination of DOCRs has been a long-standing research topic in power system protection [4, 5].

Researchers have proposed numerous techniques to deal with the challenges associated with DOCR coordination. The coordination of directional overcurrent relays (DOCRs) remains a challenging problem in power system protection due to the nonlinear characteristics of relay curves and the large number of coordination constraints, such as plug setting, time multiplier setting, and discrimination margins. Traditional mathematical programming techniques often struggle to provide robust solutions for these highly constrained and multimodal problems. To overcome such limitations, metaheuristic algorithms have been increasingly applied in recent years. Approaches based on hybrid Biogeography-Based Optimization with PSO have shown promising results in improving coordination efficiency [6]. Hybrid PSO with simulated annealing was reported to enhance global search ability and accelerate convergence compared to classical PSO [7, 8]. Other studies have focused on practical constraints such as inrush currents, distributed generation, and protection of renewable-integrated systems, where metaheuristic algorithms demonstrated superior adaptability [9, 10]. Moreover, swarm intelligence methods beyond classical PSO, such as Grey Wolf Optimization (GWO), have been tested for DOCR coordination and were found to be competitive alternatives [11]. Several comparative investigations also highlighted the advantages of using PSO and its variants for the joint coordination of overcurrent and distance relays [12]. Advanced PSO variants like time-varying acceleration coefficient PSO (PSO-TVAC) were proposed to handle series-compensated networks with improved convergence and solution quality. Recent work has also demonstrated that pure PSO can effectively optimize relay settings when integrated with updated IEC characteristics and practical operating conditions [13]. In addition, hybrid protection schemes combined with relay coordination strategies are being developed to strengthen distribution networks with high penetration of renewables [10]. Finally, comprehensive studies on distribution system

relay coordination continue to emphasize that metaheuristic approaches are more effective than conventional methods, offering higher robustness and efficiency in solving the complex DOCR coordination problem [14].

Another critical factor in DOCR coordination is the selection of the relay characteristic curve. The International Electrotechnical Commission (IEC) defines several inverse-time characteristics, including Standard Inverse (SI), Very Inverse (VI), and Extremely Inverse (EI) [15]. These characteristics directly influence the operating times of relays and, consequently, the overall coordination performance. To analyze the effect of these characteristics on the optimization outcome, the method introduced in [16] is adopted as the reference framework. In this approach, hybrid PSO and heuristic PSO are combined to minimize the total operating time of directional overcurrent relays, achieving up to 74.9% faster operation compared with conventional techniques. However, this study was limited to DOCR coordination with the IEC SI curve, and the impact of the proposed method on other IEC curves was not investigated. Therefore, this paper presents a comparative analysis between conventional PSO and the proposed hybrid PSO approach for optimal DOCR coordination in the IEEE 6-bus test system. The study further examines how different IEC characteristics (SI, VI, and EI) influence coordination results. The objective function is formulated to minimize the total relay operating time while maintaining selectivity constraints. Simulation results demonstrate the differences between PSO and hybrid PSO in terms of convergence behavior, solution quality, and robustness under various relay characteristics.

The remainder of this paper is organized as follows: Section 2 presents the mathematical formulation of the coordination problem, including the objective function and selectivity constraints. Section 3 introduces the PSO and hybrid PSO algorithms employed in this study. Section 4 describes the IEEE 6-bus test system and the simulation setup. Section 5 discusses the comparative results and provides an analysis of the impact of different IEC characteristic curves. Finally, Section 6 concludes the paper and highlights directions for future research.

2. PROBLEM FORMULATION

The problem formulation in this study is based on [16], with certain modifications to incorporate different IEC characteristic curves."

A. OBJECTIVE FUNCTION AND CONSTRAINS

The primary objective of this optimization is to minimize the operation time of coordinated DOCRs under system fault conditions. The operating time of each relay, according to the IEC characteristic curve, is formulated as follows:

$$T_i = \frac{TMS_i \times a}{\left(\frac{I_{fi}}{I_{pi}}\right)^b - 1} \quad (1)$$

Where, T_i is the operation time, TMS_i is the time multiplier setting, I_{fi} is the fault current and I_{pi} is the pickup setting of i -th relay. The coefficients a and b are defined by the IEC standard, as presented in Table 1 [15].

Table 1- Coefficient of different IEC OC curve

Curve type	a Coefficient	b Coefficient
SI	0.14	0.02
VI	13.5	1
EI	80	2

Base on the defined operation times, the objective function of optimization is defined by (F) [8].

$$F = \sum_{i=1}^N \omega_i T_i \quad (2)$$

Where, ω_i is the weight of the i -th relay's operation time and N is the total number of relays. The objective function value depends on the operation times of the primary relays under different fault scenarios. Based on Equation (1), this operation time is expressed as a function of the Time Multiplier Setting (TMS) and the Plug Setting (PS) of each relay, which serve as the optimization variables. The relation of pickup current and PS is given in Equation (3).

$$I_{pi} = PS_i \times CTR_i \quad (3)$$

Where, CTR is current transformer ratio of each relay. In the current optimization, the PS is kept constant to simplify the process, meaning that only the TMS is treated as the optimization variable. Under this assumption, the coordination problem can be formulated as a linear programming model.

Due to optimization issue and operation time, restrictions are performed. The first one in considered for TMS given in Equation (4), where TMS range is considered between 0.1 to 1.2 [16].

$$TMS_{i,min} \leq TMS_i \leq TMS_{i,max} \quad (4)$$

Although the main objective of the optimization is to minimize the operating time of primary relays, this process must also consider the coordination of backup relays with the primary relay as an additional constraint. This limitation is imposed through a crucial parameter known as the coordination time interval (CTI), given in Equation (5).

$$T_{Backup} - T_{Primary} \leq CTI \quad (5)$$

The Coordination Time Interval (CTI) typically ranges between 0.1 and 0.3 seconds, depending on the specifications of the network and associated equipment. The operating times of the main and backup relays are calculated as follows, where the indices B and P represent the backup and primary relays, respectively [17].

$$T_{Backup} = \frac{TMS_B \times a}{\left(\frac{I_{fB}}{PS_B \times CTR_B}\right)^b - 1} \quad (6)$$

$$T_{Primary} = \frac{TMS_P \times a}{\left(\frac{I_{fP}}{PS_P \times CTR_P}\right)^b - 1} \quad (7)$$

B. HPSO ALGORITHM EXPLANATION

The reference framework [16] adopted in the current study combines Particle Swarm Optimization (PSO) with Simulated Annealing (SA) to propose a more efficient approach for optimizing the coordination of directional protection relays with the name of HPSO. The proposed method, which integrates Particle Swarm Optimization (PSO) with Simulated Annealing (SA), is illustrated in the flowchart presented in Figure 1.

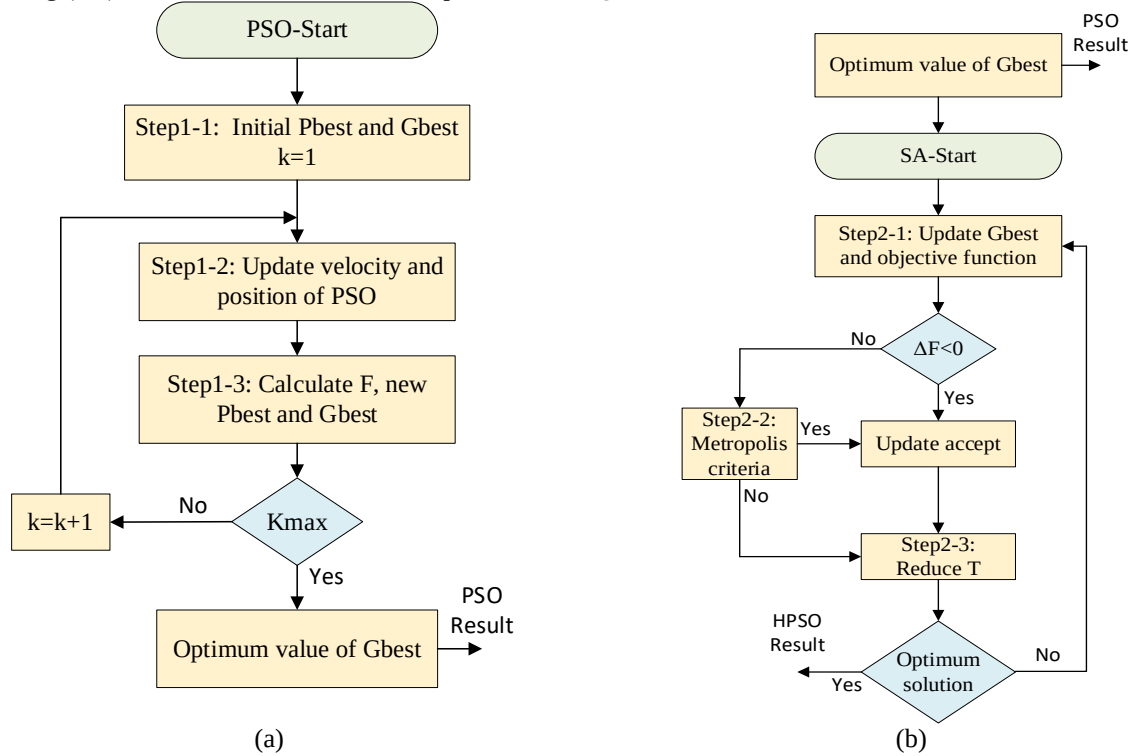


Figure 1. HPSO algorithm flowchart: (a) PSO algorithm, (b) SA algorithm

Particle Swarm Optimization (PSO) is a population-based optimization technique inspired by the social behavior of birds flocking or fish schooling, where a group of candidate solutions (particles) move through the search space by adjusting their positions based on their own best experience and the best experience of the swarm, to converge toward an optimal solution. As shown in Figure 1.a, the optimization algorithm is started with step 1-1 is included following steps:

- Determine initial position and velocity of particles:

$$X_j^1 = X_{min} + u_j(X_{max} - X_{min}) \quad (8)$$

$$V_j^1 = rand(-V, V) \text{ where } V = 0.2(X_{max} - X_{min}) \quad (9)$$

Where, X_j^1 denotes vector of optimization variables initial position for the j-th particle, u_j represents the random vector associated with the j-th particle, containing values uniformly distributed between 0 and 1 and represents vector of particles initial velocity.

- Determine initial personal best (Pbest) and global best (Gbest):

$$Pbest_j = X_j^1 \quad (10)$$

$$Gbest = \arg - \min(Pbest_j) \text{ for all particles} \quad (11)$$

The PSO algorithm proceeds by updating the particle velocity's and position's in each iteration, following the steps outlined in 1-2:

$$V_j^{k+1} = \omega V_j^k + c_1 r_1 (Pbest_j - X_j^k) + c_2 r_2 (Gbest - X_j^k) \quad (12)$$

$$X_j^{k+1} = X_j^k + V_j^{k+1} \quad (13)$$

Where, ω is the inertia with value between 0.1 to 0.9, c_1 and c_2 are acceleration factors with value around 2 and r_1 and r_2 are random value between 0 to 1 [16].

Then Pbest and Gbest are updated in step 1-3:

$$\text{if } F(X_j^{k+1}) < Pbest_j \text{ then } Pbest_j = X_j^{k+1} \quad (14)$$

$$\text{if } F(Pbest_j) < F(Gbest) \text{ then } Gbest = Pbest_j \quad (15)$$

This procedure is repeated until the stopping criterion is met, after which the PSO results are refined using the SA method. Simulated Annealing (SA) is a probabilistic optimization algorithm inspired by the annealing process in metallurgy, where a material is heated and then gradually cooled to reach a state of minimum energy. In optimization, SA explores the search space by occasionally accepting worse solutions with a certain probability, thereby helping the algorithm escape local minima and converge toward a global optimum. The optimum result of PSO is used as the initial value of SA is shown in Figure 1.b. SA starts with selecting a new value of Gbest and calculating the new value of objective function in step 2-1.

$$Gbest^{new} = Gbest^{old} + rand(-0.05, 0.05) \quad (16)$$

$$\Delta F = F(Gbest^{new}) - F(Gbest^{old}) \quad (17)$$

Now, if $\Delta F < 0$, it means that the new Gbest provides an improved solution, and thus the new value of Gbest is accepted. Otherwise, the Metropolis criterion [18] is applied in step 2-2. In this criterion, the probability function P for accepting the modification is calculated using Equation (18).

$$P = \exp\left(-\frac{\Delta F}{T}\right) \quad (18)$$

Here, T denotes the temperature in the SA procedure. The probability P is then compared with a random value r between 0 and 1. If $P > r$, the modification is accepted; otherwise, only the temperature T is updated with Equation (19) in step 2-3.

$$T^{new} = \alpha T^{old} \quad (19)$$

Here, α denotes the cooling factor. Based on published research, the initial temperature and the cooling factor of the SA procedure are considered as 1 and 0.99, respectively [16]. The SA procedure is repeated until the stopping criterion is met, which means that either the temperature reaches zero or the solution converges to its optimum value.

3. PROBLEM SIMULATION ON THE IEEE 6-BUS SYSTEM

To compare the effect of using different IEC operation curves on the optimization results of directional overcurrent relays with the PSO and HPSO methods, the IEEE 6-bus system shown in Figure 2 is used. The directional overcurrent relays are labeled R1 through R6. For the optimization process, the lower and upper limits of the TMS are set to 0.1 and 1.1, respectively, while the CTI value is considered to be 0.3 s and PS has a fixed value equal to 1.5. The CTR and pickup settings of the relays, along with the short-circuit currents of the primary and backup relays, are presented in Tables 2 and 3, respectively [16].

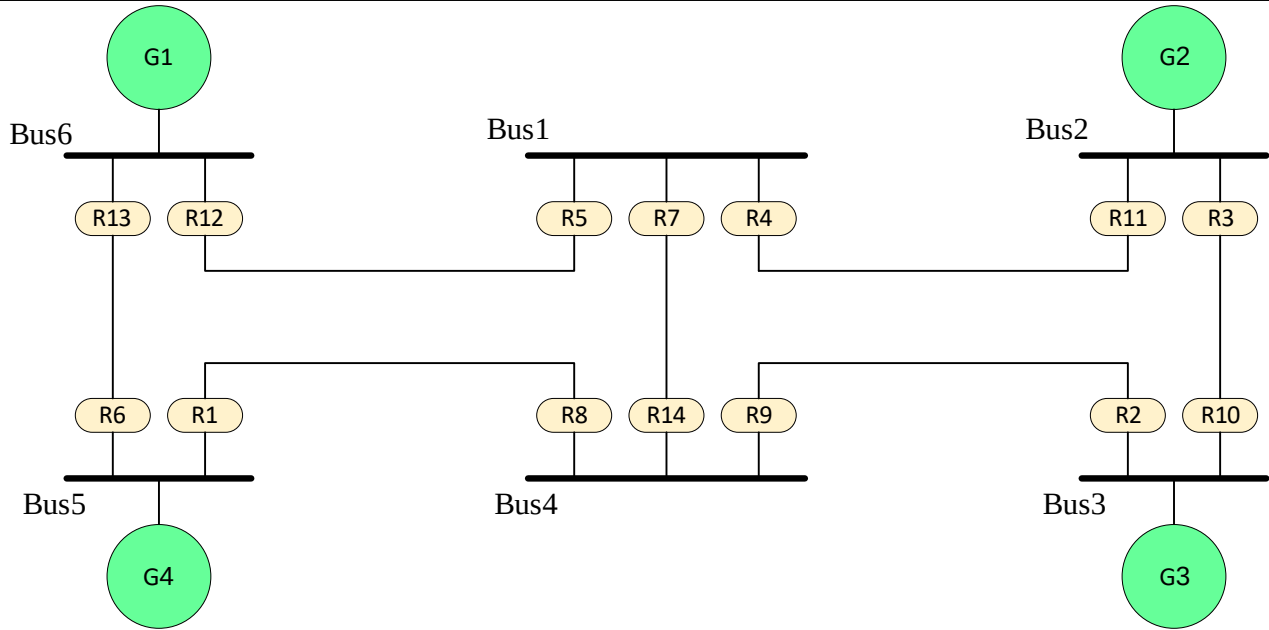


Figure 2. IEEE 6-bus system

Table 2. CTR and pickup setting of IEEE 6-bus system

Relay	CTR
R1, R6, R13	1200/5
R2, R3, R4, R5, R7, R8, R9, R11, R12, R14	200/5
R10	600/5

Table 3. Short circuit currents of IEEE 6-bus system

Primary relay	Fault current (A)	Secondary relay	Fault current (A)	Primary relay	Fault current (A)	Secondary relay	Fault current (A)
R9	6072	R1	4589	R6	18338	R8	767
R14	5457	R1	4589	R10	4077	R9	639
R8	2351	R2	868	R11	30939	R10	945.5
R14	5457	R2	868	R5	2838	R11	1074
R2	4803	R3	1365	R7	4496	R11	1074
R3	30547	R4	552.8	R4	5186	R12	3422
R13	17821	R5	977	R7	4496	R12	3422
R12	17705	R6	861	R1	18172	R13	601
R8	2351	R7	1483	R4	5186	R14	1764
R9	6072	R7	1483	R5	2838	R14	1764

The convergence of the PSO and HPSO methods using the IEC-SI curve for protection relays is shown in Figure 3. The results indicate that PSO fails to reach the optimum solution even after around 100 iterations, whereas HPSO, with the aid of SA, successfully improves the solution in fewer than 100 iterations.

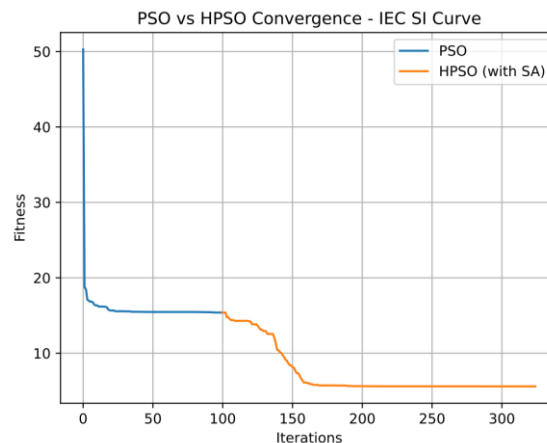


Figure 3. PSO and HPSO convergence for relays with IEC-SI curve

Table 4. PSO and HPSO result for relays coordination with IEC-SI curve

	TMS														O.F
	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	
PSO	0.374	0.739	0.872	0.205	0.219	0.970	0.342	0.330	0.150	0.230	0.475	0.866	0.159	0.264	15.39
HPSO	0.247	0.123	0.147	0.100	0.100	0.100	0.159	0.100	0.100	0.120	0.145	0.260	0.100	0.168	5.59

Table 4 also clearly demonstrates that HPSO improves the optimization results, leading to faster fault clearing in the system. The convergence of HPSO and PSO methods for optimizing the coordination of directional overcurrent relays with IEC-VI and IEC-EI characteristics is presented in Figure 4. The results indicate that PSO faces similar challenges in finding the optimal solution when the IEC-VI and IEC-EI curves are used, while SA improves the results in both cases. Table 5 and Table 6 present the TMS and objective function values obtained from PSO and HPSO for relay coordination with IEC-VI and IEC-EI, respectively. The findings show that, in both cases, optimization with HPSO results in lower objective function values.

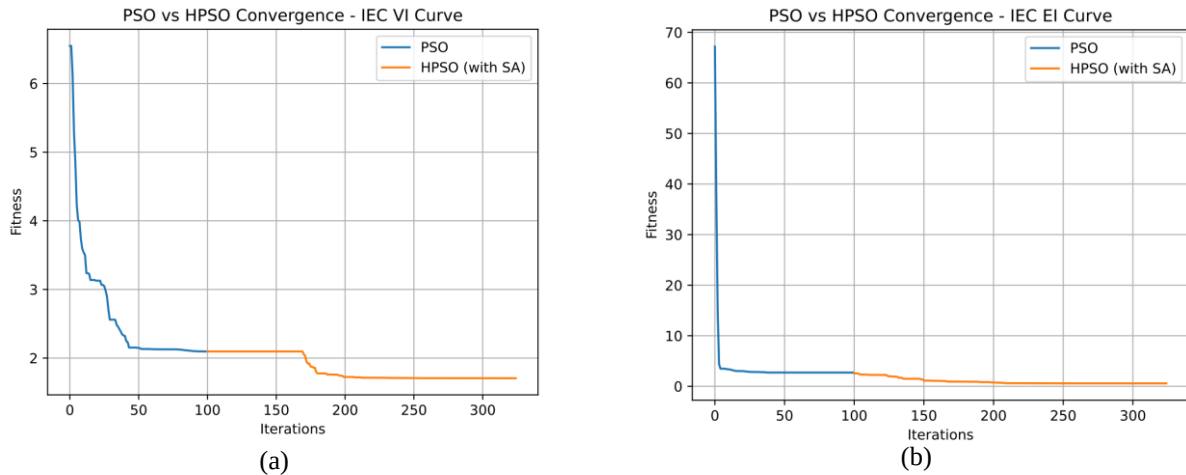


Figure 4. PSO and HPSO convergence for relays with: (a) IEC-VI curve, (b) IEC-EI curve

Table 5. PSO and HPSO result for relays coordination with IEC-VI curve

	TMS														O.F
	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	
PSO	1.100	0.100	0.176	0.100	0.118	0.100	0.174	0.100	0.100	0.100	0.116	1.100	0.100	0.211	2.096
HPSO	0.370	0.100	0.129	0.100	0.100	0.100	0.174	0.100	0.100	0.100	0.111	0.425	0.100	0.200	1.706

Table 6. PSO and HPSO result for relays coordination with IEC-EI curve

	TMS														O.F
	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	
PSO	1.100	0.170	0.136	0.163	0.100	0.264	0.524	0.976	1.100	1.100	1.100	1.100	1.100	0.237	2.712
HPSO	0.680	0.100	0.127	0.100	0.100	0.100	0.179	0.100	0.100	0.100	0.100	0.862	0.100	0.237	0.575

5. CONCLUSIONS

This study investigated the optimal coordination of Directional Overcurrent Relays (DOCRs) in the IEEE 6-bus test system using PSO and HPSO under different IEC inverse-time characteristics. The results clearly show that while conventional PSO provides acceptable solutions, it struggles to reach global optima within a reasonable number of iterations. In contrast, the proposed HPSO approach, by integrating SA with PSO, successfully improves convergence speed, avoids premature stagnation, and produces significantly lower objective function values. Comparative analysis across SI, VI, and EI curves highlights that HPSO achieves faster fault clearance and more reliable coordination than PSO alone. These improvements are 63%, 18%, and 78% for relays with IEC-SI, IEC-VI, and IEC-EI characteristics, respectively. These findings highlight the potential of hybrid metaheuristic approaches in addressing the nonlinear and highly constrained nature of relay coordination problems under different operational characteristics. Future research could extend this work to larger and more complex power systems, incorporate renewable energy sources, and explore adaptive hybridization strategies for real-time protection applications. Future research could extend this work in several directions. First, applying the method to larger IEEE test systems or real-world networks would provide further insight into scalability and robustness. Second, integrating renewable energy sources such as solar and wind would allow the evaluation of relay coordination under fluctuating fault levels. Third, adaptive and real-time coordination strategies could be developed to automatically adjust relay settings with changing network conditions. Additionally, multi-objective optimization frameworks could be explored to simultaneously address operating time, reliability, and sensitivity. Finally,

benchmarking against other hybrid metaheuristics and validating results through hardware-in-the-loop or real-time digital simulation would enhance the practical applicability of the proposed approach.

6. REFERENCES

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